Hritik Saynganthone

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OBJECTIVE

To obtain a full-time position in software development, located either in-person/hybrid in and around Rochester, Syracuse, or Buffalo, NY. Alternatively, anywhere remote. Available starting immediately.

EDUCATION

Rochester Institute of Technology

Rochester, NY

M.S. in Computer Science, Minor in Mathematics — May 2026

Member of the *Document and Pattern Recognition Lab*

Relevant Courses: *Machine Learning, Foundations of Computer Vision, Global Illumination, Introduction to Software Engineering, Concepts of Parallel and Distributed Systems, Computational Complexity*

WORK EXPERIENCE

Solü Technology Partners

Victor, NY

Software Development Intern — Jan. '23 - Aug. '23

- Worked with an established code base with other interns, utilizing Agile Scrum methodologies within the Atlassian suite of tools.
- Utilized several JVM-based technologies for backend development of new web application features.
- Interfaced with both MySQL and MongoDB databases for user and application data, respectively.
- Created several new frontend features, applying MVC design patterns and component-based architectures using Angular.
- Verified and produced several API endpoints using Postman, to be hit in various stages of the application lifecycle.
- Unified and simplified software infrastructure by completely containerizing the entire development, testing, and production environments with Docker Compose and AWS ECR/ECS.
- Introduced the use of automated tested via AWS CodePipeline for simultaneous backend, frontend, API, and system integration tests.
- Technologies used: **Java**, **Kotlin**, **OOP**, **Maven**, **Gradle**, **Spring Boot**, **RESTful APIs**, **Mongock**, **MongoDB**, **MySQL**, **Angular**, **MVC**, **TypeScript**, **HTML**, **CSS**, **Docker**, **Docker Compose**, **Microservices Architecture**, **AWS CodePipeline**, **AWS ECR/ECS**, **AWS EC2**, **CI/CD**, **Python**, **Postman**, **Swagger**, **Selenium**, **Git**, **Trello**, **Agile Scrum**, **Confluence**, **BitBucket**.

Software Development Intern — May '24 - Dec. '24

- Continued the development and upkeep of previously implemented systems and features.
- Led the development of the Solü Application Service Tool, a Python-based command line tool that can be used by developers to quick instantiate different environments on any system they are using.
- Initialized the development of a multi-tenant system, allowing a different backend system to be loaded with different data without needing a completely separate environment.
- Technologies used: **Python**, **Bash**, **Spring Boot**, **SQL**, **Docker**.

Software Development Intern — May '25 - Aug. '25

- Maintained the previously developed web application as the sole developer on the project.
- Implemented an MVP Retrieval Augmented Generation system using LibreChat, such that company data could be stored in a vector database and an LLM could be queried for information therein.
- Technologies used: **Python**, **Azure AI Foundry**, **Azure CosmosDB**, **LibreChat**, **ChatGPT**.

WORK EXPERIENCE (CONT.)

Kate Gleason College of Engineering — Rochester, NY

Research Assistant — Jan. ’22 - May ‘22

- Designed and developed the Hybrid Learning Optimization Solver Architecture, used to wire several mathematical optimizers together for finding approximations to NP-hard problems.
- Instituted a client-server oriented API to interface with multiple solvers, including both traditional algorithms and machine learning algorithms.
- Technologies used: **C#**, **Java**, **Python**, **.NET**, **Apache Kafka**, **FastAPI**, **Git**.

ADDITIONAL PROJECTS

Master’s Thesis — Jan. ’24 - May ’26

- Conducted novel research within the machine learning field of Optical Music Recognition, submitted for both Master’s Thesis approval and the International Journal on Document Analysis and Recognition.
- Explored high-entropy alternatives to the Connectionist Temporal Classification (CTC) loss function, utilizing convolutional recurrent neural networks (CRNNs).
- Technologies used: **Python**, **Conda**, **Virtualenv**, **PyTorch**, **TensorFlow**, **CUDA**, **Music21**, **NumPy**, **OpenCV**, **Matplotlib**, **Git**, **Gitlab**, **Github**, **Linux**, **LaTeX**, **Slurm**.

Personal Website Built in Django — June ‘26

- Used a lack of personal website as a reason to learn Django and get better at raw HTML/CSS.
- Utilized Docker Compose and Cloudflare tunnels to create completely self-host the website on my own personal server.
- Technologies used: **Python**, **Django**, **PostgreSQL**, **Docker Compose**, **Docker**, **Gunicorn**, **Virtualenv**, **Makefile**, **Nginx**, **Cloudflare Tunnels**, **HTML**, **CSS**, **Git**.

Airspace Sectorization via Voronoi Diagrams — Aug. ’25 - Dec. ’25

- Developed an algorithm to sectorize a given airspace using a voronoi diagram based on restrictions simulated via a genetic algorithm.
- Technologies used: **Python**, **Genetic Algorithms**, **Voronoi Diagrams**, **OpenSky**, **FastAPI**, **NumPy**, **Git**, **GitHub**.

From Scratch Raytracing Engine — Jan. ’25 - May. ’25

- Developed multithreaded raytracing engine that can render simple objects within 3D space.
- Technologies used: **C#**, **Thread Pools**, **Git**, **GitHub**.

Full Stack E-Store Web Application — Jan. ’22 - May ’22

- Utilized Java Spring Boot & Angular in order to uphold key object-oriented design principles.
- Employed the full Agile development cycle with a team of software developers from start to finish.
- Technologies used: **Java**, **Spring Boot**, **RESTful APIs**, **Angular**, **Agile Scrum**, **Trello**, **SonarQube**, **Git**, **GitHub**.

Depth First Search Sudoku Solver — July ‘21

- Independently created a Java interface and class file to solve DFS-compliant puzzles.
- Formulated and executed a way to take in 9x9 sudoku puzzles and output a solution as a command line application.
- Technologies used: **Java**, **Git**, **GitHub**.

EXTRACURRICULAR ACTIVITIES

RIT Game Symphony Orchestra — Member, Wind Section Lead — Aug. '20 - May ’22, May ’22 - May ’23
RIT Game Symphony Orchestra — Assistant Conductor, Conductor — May. '23 - May ’25, May ’25 - May ’26